

Estimating Wind Velocity from Limited Aircraft Flight Data

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Abstract—Maximum likelihood (ML) and maximum *a posteriori* (MAP) estimators are described for the estimation of wind velocity using position, and airspeed measurements from flight data recordings. The estimators are demonstrated using simulated flight data and experimental flight data from a high-altitude sailplane flight.

Keywords—Maximum likelihood, maximum *a posteriori*, wind velocity, aircraft, sailplane.

I. INTRODUCTION

Atmospheric mountain waves, or lee waves, are a form of atmospheric gravity wave that are produced by air flow over elevated terrain in a stable, stratified atmosphere [1], [2], [3]. Atmospheric gravity waves arise from buoyancy forces when an air parcel is displaced vertically from its equilibrium position, leading to harmonic motion. Lee waves are of significant importance in meteorology. They can influence the vertical structure of wind speed and temperature fields, cause fluctuations in wind speeds in the lower atmosphere, and can affect ozone concentration [4], [5], [6]. Turbulence associated with mountain waves can produce strong vertical velocities, particularly down drafts, that can be an aviation hazard [7], [8]. The spatial structure of mountain waves is therefore of considerable interest in meteorology and aviation.

Our interest here is that of estimating the wind velocity in lee waves using simple data collected during an aircraft flight. Wind fields in lee waves generally vary slowly in space and time. We utilise this slowly varying characteristic to develop a Bayesian statistical framework for estimating wind velocities when the available flight data are limited. This approach extends a previously developed deterministic method [9], [10]. Sailplane flights are a good source of flight data due to their relatively simple and well-characterised aerodynamics. Sailplane pilots make use of mountain lee waves to generate lift and gain altitude, and data are routinely recorded during flights, offering considerable potential for studying mountain wave structures.

In this paper we focus on estimating horizontal wind velocity, and the methods can be easily extended to 3 dimensional. The flight data and measurement model are

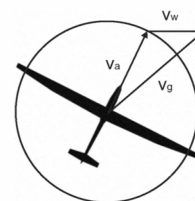


Figure 1. The aircraft ground velocity $\mathbf{v}^g$ is the vector sum of the aircraft velocity relative to the airmass $\mathbf{v}^a$ and the wind velocity $\mathbf{v}^w$.

described in the next section. In Section III, a ML estimator is described and is evaluated by simulation. A MAP estimator is described in Section IV and also evaluated by simulation. Application of these methods to sailplane data recorded during a flight in atmospheric wave in the lee of the Sierra Nevada mountain range is presented in Section V. Concluding remarks are made in Section VI.

II. FLIGHT DATA AND MODEL

Consider an aircraft flying through a region of slowly varying wind speed and direction in time and position. The problem we consider is that of estimating the horizontal wind speed and direction along the flight path from simple flight data measured by on-board instruments. Here we consider the case in which the available data are ground velocity derived from GPS position measurements and air-speed (speed of the aircraft relative to the local airmass). We use the fundamental relationship

$$\mathbf{v}^g = \mathbf{v}^a + \mathbf{v}^w, \quad (1)$$

where the superscripts g , a , and w denote "ground", "air" and "wind", respectively (Fig. 1). As the angle of air velocity is unknown, there is a one-parameter family of solutions for the wind velocity.

The three velocities represent six parameters, four of which are independent. We use the ground and wind velocities as a non-redundant parameter set that describes the "state" at each time-step. The state is then given by

$$(v_{xi}^g, v_{yi}^g, v_{xi}^w, v_{yi}^w)^T, \quad (2)$$

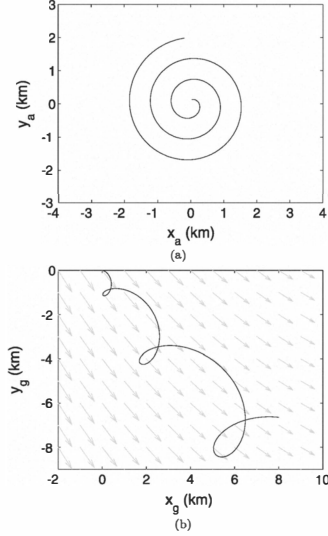


Figure 2. Flight path for the simulated data (a) relative to the air and (b) relative to the ground. The arrows in (b) show the vector wind field.

where the subscripts x (East) and y (North) denote the Cartesian components and i denotes the i^{th} time-step.

The data consist of the ground velocity and airspeed which are measured at each time-step, and are denoted by

$$(\tilde{v}_{xi}^g, \tilde{v}_{yi}^g, \tilde{v}_i^a)^T, \quad (3)$$

where $(\tilde{v}_{xi}^g, \tilde{v}_{yi}^g)$, and $\tilde{v}_i^a$ are the measured ground velocity components, and airspeed, respectively. The instrumentations used for measurement are checked and calibrated regularly to ensure that the measurement is within acceptable degrees of accuracy. For simplicity, we model the data as true data with additive Gaussian white noise. The measured ground velocity components are related to the true ground velocity components (state parameters) by

$$\tilde{v}_{xi}^g = v_{xi}^g + n_{xi}^g \quad (4)$$

$$\tilde{v}_{yi}^g = v_{yi}^g + n_{yi}^g, \quad (5)$$

where n_{xi}^g, n_{yi}^g are errors that are modelled as zero-mean, white Gaussian noise with variance σ_g^2 . For a sailplane, the glide slope is shallow and the airspeed is very close to its horizontal component so that

$$\tilde{v}_i^a = \sqrt{(v_{xi}^g - v_{xi}^w)^2 + (v_{yi}^g - v_{yi}^w)^2} + n_i^a \quad (6)$$

where n_i^a are the airspeed errors also modelled as zero-mean, white Gaussian noise with variance σ_a^2 .

III. MAXIMUM LIKELIHOOD ESTIMATOR

Consider first a ML estimator of the wind velocity. At a single time step, the number of parameters is greater than the number of data, which leads to a non-unique, one parameter family of solutions for the wind velocity (Fig. 1). We have

shown previously that the ambiguity is resolved for data at multiple time steps with a constant wind velocity, if the ground velocities are different [9], [10].

Taking a statistical estimation approach, using data from a number of contiguous time-steps with the assumption of constant wind velocity a ML estimator can be developed as follows. We define the state and data vectors, $\mathbf{s}_i$ and $\mathbf{d}_i$, respectively, at time-step i by the values within a temporal window $[i - N, i + N]$, i.e.,

$$\mathbf{s}_i = (v_{x,i-N}^g, v_{y,i-N}^g, \dots, v_{x,i+N}^g, v_{y,i+N}^g, v_{xi}^w, v_{yi}^w)^T, \quad (7)$$

$$\mathbf{d}_i^a = (\tilde{v}_{x,i-N}^g, \tilde{v}_{y,i-N}^g, \tilde{v}_{i-N}^a, \dots, \tilde{v}_{x,i+N}^g, \tilde{v}_{y,i+N}^g, \tilde{v}_{i+N}^a)^T. \quad (8)$$

Using the assumption of Gaussian noise, the likelihood function is

$$p(\mathbf{d}_i|\mathbf{s}_i) = K_1 \prod_{j=-N}^N \exp \left(- \left[(\tilde{v}_{x,i+j}^g - v_{x,i+j}^g)^2 + (\tilde{v}_{y,i+j}^g - v_{y,i+j}^g)^2 \right] / 2\sigma_g^2 - \left[\tilde{v}_{i+j}^a - \sqrt{(v_{x,i+j}^g - v_{xi}^w)^2 + (v_{y,i+j}^g - v_{yi}^w)^2} \right]^2 / 2\sigma_a^2 \right). \quad (9)$$

As Eq.9 is a Gaussian likelihood function, K_1 is chosen to be inversely proportional to $\sigma_a\sigma_g$. The variances σ_a and σ_g are chosen based on empirical data and sensitivity analysis. The ML estimate of the state is then given by

$$\hat{\mathbf{s}}_i^{\text{ML}} = \underset{\mathbf{s}_i}{\text{argmax}} \{p(\mathbf{d}_i|\mathbf{s}_i)\}, \quad (10)$$

which gives an estimate of the wind velocity. The performance of the ML estimator is illustrated using simulated flight data. The wind field used varies linearly in magnitude and direction as a function of x as shown in Fig. 2. The magnitude varies from 35 m/s to 25 m/s and the direction varies from 305° to 325° between $x = 0$ km and $x = 8$ km. Following conventional meteorological notation, we quote the wind direction as the direction the wind is coming *from*. The aircraft performs a spiral manoeuvre within the airmass for 400 s at a constant airspeed of 50 m/s; the start radius is 100 m and the radius increases linearly with time to an end radius of 2 km. Measurements of the ground velocity and airspeed are taken at 1 s intervals and are corrupted by noise with standard deviations of 2 m/s and 1 m/s, respectively.

The flight paths relative to the air and relative to the ground are shown in Fig. 2. The values of σ_g, σ_a used are 2 m/s and 2 m/s, respectively. The ML estimates were calculated as described above using $N = 3$ and $N = 9$. The errors in the estimated wind speed and direction are plotted versus time for 5 noise realisations in Fig. 3 (a) and (b) for $N = 3$ and in Fig. 3 (c) and (d) for $N = 9$. The rms errors

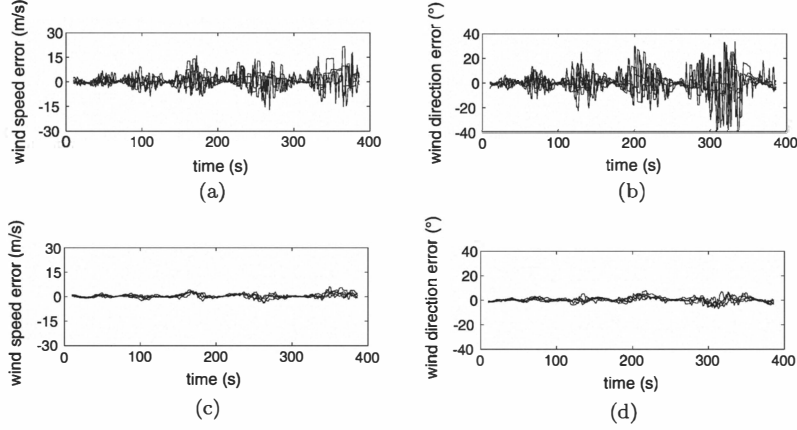


Figure 3. Errors in the ML wind speed and direction estimates for 5 different noise realisations. The estimates were obtained using position and airspeed data, using (a)(b) sets of 7 subsequent samples ($N = 3$), and (c)(d) sets of 19 subsequent samples ($N = 9$).

are 4.2 m/s and 8.0° for $N = 3$, and 1.3 m/s and 1.9° for $N = 9$. The estimates are quite good for $N = 9$ but the errors are larger for $N = 3$. This is due to more averaging for $N = 9$, but also because the larger temporal window captures a wider range of ground track directions.

IV. MAXIMUM A POSTERIORI ESTIMATOR

As shown in the simulations presented in the previous section, the sensitivity of the ML estimators to noise in data can be reduced by using more data at additional time-steps. However, the assumption of a constant wind velocity over the $2N + 1$ time steps becomes less valid as N increases. Increasing N does not therefore guarantee a reduction in error. The assumption of a constant wind velocity can be relaxed by incorporating prior information that the wind field is a slowly varying function of space and time. This is implemented by constraining the difference between the wind velocity estimate and previous estimates. For simplicity, we consider differences in time only, as the separations in position and time are loosely coupled. Since contiguous wind velocity estimates will already be correlated, previous estimates spaced by Q time steps are used. The prior distribution is then given by

$$p(\mathbf{s}_i) = K_2 \prod_{k=1}^M \exp \left(-a_{Qk} \left[(v_{xi}^w - \hat{v}_{x,i-Qk}^w)^2 + (v_{yi}^w - \hat{v}_{y,i-Qk}^w)^2 \right] / 2\sigma_w^2 \right) \quad (11)$$

where we use a window of length MQ time steps, a_k is a weight that is a decreasing function of k to express the fact that estimates spaced further in time are less correlated, and σ_w^2 is a scaled variance. For a random walk ground track in a time-invariant wind field, a reasonable choice for a_k is $a_k = k^{-1}$, and for a straight track $a_k = k^{-2}$. We have found that the former works well. As Eq.11 is a Gaussian prior,

K_2 is chosen to be inversely proportional to σ_w . The prior distribution is incorporated into a maximum *a posteriori* (MAP) estimate as

$$\begin{aligned} \hat{\mathbf{s}}_i^{\text{MAP}} &= \underset{\mathbf{s}_i}{\text{argmax}} \{p(\mathbf{s}_i | \mathbf{d}_i^a)\} \\ &= \underset{\mathbf{s}_i}{\text{argmax}} \{p(\mathbf{d}_i^a | \mathbf{s}_i) p(\mathbf{s}_i)\}. \end{aligned} \quad (12)$$

The performance of the MAP estimator is illustrated using the simulated data described in Section III and the results are shown in Fig. 4. The MAP estimates exhibit a slowly varying error due to being constrained to the previous estimates, but this error is relatively small. Again we used $\sigma_g = 2$ m/s and $\sigma_a = 2$ m/s. The wind speed is assumed to vary slowly with time at no more than 2 times σ_w per second most of the time, and a subjective choice of σ_w was 5 m/s. And we used $a_k = k^{-1}$, $N = 3$, $M = 3$, and $Q = 3$. These values of N , M , and Q constrain the true wind velocity to be constant over a window of length 7 as in Fig. 3(a)(b), and constrain it by the prior over a time interval of length 9, which is the same as the half-width of the window length in Fig. 3(c)(d). The rms errors are 1.4 m/s and 3.1° . These errors are smaller than for the ML estimator for $N = 3$ but slightly larger than for $N = 9$. The MAP estimator is expected to be advantageous over ML if the wind field varies more rapidly.

V. EXPERIMENTAL RESULTS

The methods described above were applied to data recorded during a sailplane flight in mountain waves in the lee of the Sierra Nevada Mountain range in Southern California. The basic data consisted of GPS position and indicated airspeed. The reader is referred to [9], [10] for details of this flight and the data. The data were low-pass filtered and ground velocities and true airspeed were calculated using standard methods [9], [10].

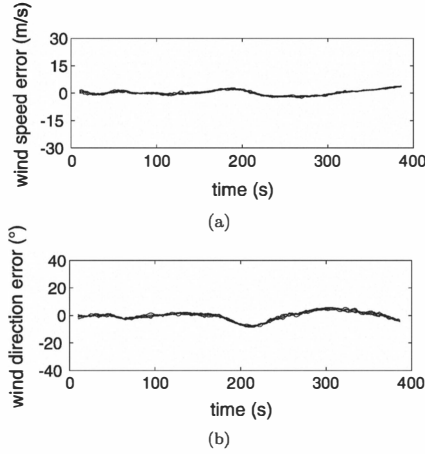


Figure 4. Errors in the MAP wind speed and direction estimates for 5 different noise realisations. The estimates were obtained using $M = 3$, $N = 3$ and $Q = 3$, using position and airspeed data.

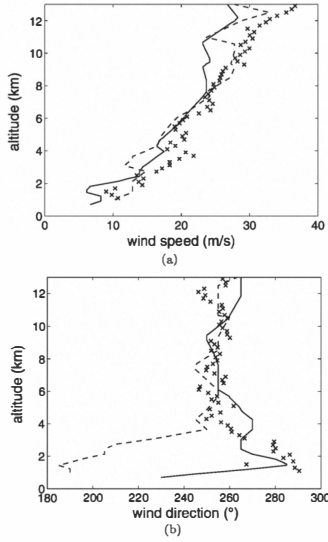


Figure 5. Estimated wind speed and direction for a sailplane flight on the Sierra Nevada mountain range. The crosses denote MAP estimates. The solid and dash lines show the Edward and Desert Rock radiosonde data, respectively.

The ground velocity and airspeed were used as data and the wind velocity estimated using the MAP estimator Eq. (12). We chose $\sigma_g = 2$ m/s, $\sigma_a = 2$ m/s, $\sigma_w = 5$ m/s, $a_k = k^{-1}$, $N = 3$, $M = 3$, and $Q = 3$. The estimated wind velocities for the whole flight were collected into altitude bins 200 m thick and averaged, and the speed and direction are shown versus altitude in Fig. 5.

For comparison purposes, the wind speeds and directions recorded by radiosonde soundings at Edwards Air Force Base, Edwards, California, (approximately 120 km south of the centre of the flight path) 9 hours before the flight, and at Desert Rock, Mercury, Nevada (approximately 180 km east

of the centre of the flight path) close to the flight time, are also shown. Considering the different locations and times of the radiosonde soundings compared to those of the flight, there is good agreement with the estimated values.

VI. CONCLUSIONS

We have described ML and MAP estimators for calculating wind velocity in atmospheric mountain waves using position and airspeed data recorded during an aircraft flight. The methods utilize an assumption of a slowly varying wind field in position and time. For slowly varying wind fields, the noise sensitivity of the ML estimator can be reduced by averaging over a longer window. Incorporating prior information into the solution and using a MAP estimator improves the performance for a varying wind field. The estimators could be further developed for application to other kinds of limited flight data.

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